

Supplementary Online Content

Writing Group for the PReVENT Investigators. Effect of a low vs intermediate tidal volumes strategy on ventilator-free days in intensive care unit patients without ARDS: a randomized clinical trial. *JAMA*. doi:10.1001/jama.2018.14280

Supplement 1. Trial protocol

This supplementary material has been provided by the authors to give readers additional information about their work.

Supplementary Online Content 1

The PReVENT writing committee; Effect of a Low versus Intermediate Tidal Volume Strategy on Ventilator-free Days in Intensive Care Unit Patients without ARDS – A Randomized Clinical Trial

Study Protocol

PROTOCOL TITLE: PRotective VENTilation in Patients without ARDS at Start of Ventilation – PreVENT, a Randomized Controlled Trial

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ABBREVIATIONS AND DEFINITIONS

ARDS	Acute Respiratory Distress Syndrome
CE	Cost-Effective
DSMB	Data Safety Monitoring Board
EAdi	Electrical Activity of the diaphragm
ICER	Incremental Cost-Effectiveness Ratios
ICU	Intensive Care Unit
ICU-AW	Intensive Care Unit – Acquired Weakness
LOS	Length Of Stay
METC	Medical Research Ethical Committee (MREC) in Dutch: <i>Medische Ethische Toetsings Commissie</i>
NAVA	Neurally Adjusted Ventilatory Assist (catheter)
PEEP	Positive End Expiratory Pressure
PBW	Predicted Body Weight
RCT	Randomized Controlled Trial
SAE	Serious Adverse Event
Sponsor	The sponsor is the party that commissions the organization or performance of the research, for example a pharmaceutical company, academic hospital, scientific organization or investigator. A party that provides funding for a study but does not commission it is not regarded as the sponsor, but referred to as a subsidizing party.
VIDD	Ventilator-Induced Diaphragm Dysfunction
VOC	Volatile Organic Compound

SUMMARY

Rationale: It is uncertain what the optimal ventilation strategy is in intensive care unit (ICU) patients without the Acute Respiratory Distress Syndrome (ARDS): ventilation with a lower tidal volume and higher respiratory rate *or* ventilation with a higher tidal volume and lower respiratory rate.

Hypothesis: Ventilation with a lower tidal volume and a higher respiratory rate increases the number of ventilator-free days and alive of patients without ARDS at onset of ventilation. As such, lower tidal volume ventilation shortens ICU length of stay and can be expected to reduce healthcare costs.

Objective: The present study aims to compare a ventilation strategy with a lower tidal volume and a higher respiratory rate with ventilation using a higher tidal volume and a lower respiratory rate in patients without ARDS at start of ventilation.

Study design: This is a national multicenter randomized controlled trial in intubated and ventilated ICU patients without ARDS, according to the Berlin definition.

Study population: Consecutive patients admitted to the ICUs of the participating centers without ARDS and an anticipated duration of ventilation of more than 24 hours.

Procedure: Patients are randomly assigned in a 1:1 ratio to lower tidal volume ventilation (4 to 6 ml/kg predicted body weight) and a higher respiratory rate, the 'lower tidal volume'-arm, or higher tidal volume ventilation (8 to 10 ml/kg predicted body weight) and a lower respiratory rate, the 'higher tidal volume'-arm. Respiratory rates are adjusted to reach a blood pH of 7.25 to 7.45.

Study endpoints: The primary endpoint is the number of ventilator-free days and alive at day 28 after ICU admission. Secondary endpoints include ICU – and hospital length of stay (LOS) and – mortality (follow up till day 90); incidence of development of ARDS, pneumonia, atelectasis, and pneumothorax; cumulative use and duration of sedatives, and neuromuscular blocking agents; incidences of ICU delirium and ICU acquired weakness; patient-ventilator asynchrony; and the need for decreasing of instrumental dead space. Finally, health care costs will be estimated from a health

systems perspective over the time horizon of the RCT. Incremental cost per ICU day avoided is the primary cost–effectiveness endpoint.

Nature and extent of the burden and risks associated with participation, benefit and group relatedness: Burden and risks of the two ventilation strategies are uncertain. Lower tidal volume ventilation could induce an increased need for sedation and increase the risk of asynchrony and respiratory muscle fatigue, while ventilation with higher tidal volumes could increase the risk of lung injury. Both could lead to a longer duration of ventilation. Collection of data causes no harm for the patients.

1. INTRODUCTION AND RATIONALE

1.1 Mechanical ventilation associated injury

Mechanical ventilation is generally seen as an invasive but safe supportive strategy in critically ill patients with respiratory failure. However, there is unequivocal evidence from both experimental and clinical studies that ventilation has a strong potential to aggravate and even initiate injury to the lungs [1, 2] and the respiratory muscles [3, 4]. Ventilation results invariably in a pattern of ventral overstretching and dorsal collapse of lung tissue, which both play a role in development of so-called ‘ventilator-induced lung injury’ (VILI). Ventilation is associated with respiratory muscle disuse and even misuse causing atrophy of diaphragmatic myofibers, which play a role in development of so-called ‘ventilator-induced diaphragm dysfunction’ (VIDD) [5].

1.2 Ventilation with lower tidal volumes in patients with ARDS

The harmful effects of the traditional use of a higher tidal volume were not recognized until 2000 when the beneficial effects of ventilation with a lower tidal volume (6 ml/kg predicted body weight, PBW) in patients with the Acute Respiratory Distress Syndrome (ARDS) were established in the landmark NHLBI ARDS Network trial [6]. Some clinicians and investigators were slow to accept these findings, but subsequent trials and a meta-analysis convincingly confirmed the reduction in mortality by using lung-protective ventilator settings in patients with ARDS [7]. Currently, lung protective ventilation is considered standard of care for ARDS (i.e., moderate or severe ARDS see paragraph 1.4) [8–11].

1.3 Uncertainty over the use of lower tidal volumes in patients without ARDS

It is uncertain whether we should also use a lower tidal volume in patients without ARDS. One meta-analysis suggested that ventilation with a lower tidal volume benefits patients without ARDS [12]. The meta-analyzed studies, however, all had methodological shortcomings and the majority of them focused on ventilation during general anesthesia for surgery, which cannot be simply generalized to other clinical situations like ventilation of critically ill patients in the ICU setting. There are several arguments against use of lower tidal volume ventilation in these patients: one argument is that a

compensatory higher respiratory rate could increase sedation needs [13], which could increase the incidence of ICU delirium [14] and ICU acquired weakness [15]; others reasons are that this strategy could promote collapse of lung tissue [16]; and the use of a higher respiratory rate could increase the risk of patient–ventilator asynchrony [17, 18]; more effort can induce more pendelluft and thereby lung injury [19], finally, an alleged side-effect of lower tidal volume ventilation is patient fatigue although there are no studies to support this. All of these could offset the beneficial effects of lower tidal volume ventilation as found in patients with ARDS. Thus, it is uncertain whether ventilation with lower tidal volume should be used routinely in *all* ICU patients. Consequently its use is not yet recommended in guidelines for ventilation of patients without ARDS resulting in remarkable and unwanted practice variation and continued use of higher tidal volumes [20, 21], as illustrated in figure 1 that shows the modus of tidal volume size in recent publications [yet unpublished data].

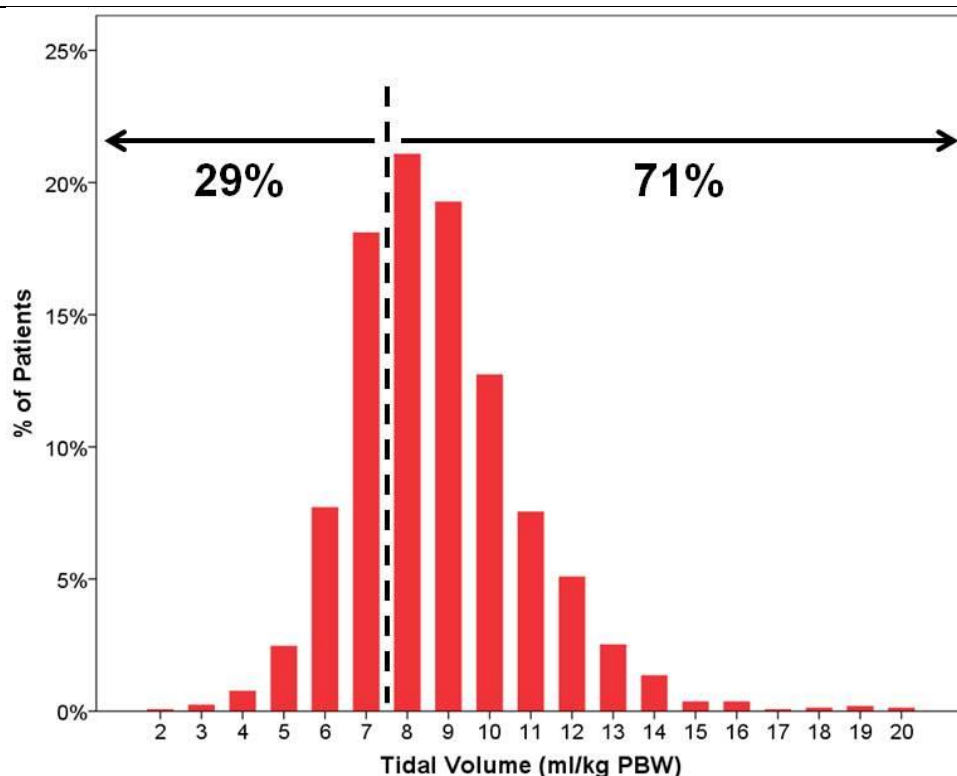


Figure 1. Histogram showing the modus of tidal volume size in ml/kg of predicted body weight in 1961 patients in four observational studies in patients not suffering from ARDS [22–25].

1.4 Need for a new randomized controlled trial of ventilation in patients without ARDS

The ICU community explicitly requests a well-powered trial comparing ventilation with a lower tidal volume to ventilation with a higher tidal volume in patients without ARDS [12, 26]. This trial should use patient-relevant and only objective outcomes, including duration of ventilation and ICU- and hospital length of stay (LOS), amongst other outcomes. The 'PRotective VENTilation in patients without ARDS at start of ventilation'-trial (PReVENT) is the first randomized controlled trial comparing lower tidal volume ventilation with higher tidal volume ventilation in patients without ARDS at onset of ventilation that recruits a sufficient number of patients to test the hypothesis that lower tidal volume ventilation benefits patients without ARDS at start of ventilation with regard to clinically relevant endpoints. Second, PReVENT is the first prospective study that will evaluate the incremental cost-effectiveness of lower tidal volume ventilation in patients without ARDS.

2. OBJECTIVES AND HYPOTHESIS

2.1 Objectives

2.1.1 Primary objective

The aim of PReVENT is to compare a ventilation strategy using lower tidal volumes and higher respiratory rates with a ventilation strategy using higher tidal volumes and a lower respiratory rate in intubated and ventilated intensive care unit (ICU) patients without Acute Respiratory Distress Syndrome (ARDS) at start of ventilation.

2.1.2 Secondary objectives

Secondary objectives are to investigate the effects of tidal volume size on ICU– and hospital LOS and – mortality, the incidence of development of ARDS, pneumonia, atelectasis, and pneumothorax, the cumulative use and duration of sedatives, and neuromuscular blocking agents, incidences of ICU delirium and ICU acquired weakness, patient–ventilator asynchrony and the need for decreasing of instrumental dead space.

2.2 Hypothesis

2.2.1 Primary hypothesis

We hypothesize ventilation with lower tidal volumes to benefit patients without ARDS at start of ventilation by increasing the number of days alive and free from ventilation at day 28. As such using a lower tidal volume instead of a higher tidal volume is expected to be a cost–effective modification of ventilation in patients without ARDS.

2.2.2 Secondary hypotheses

We hypothesize ventilation with lower tidal volumes to reduce ICU – and hospital LOS and – mortality, the incidence of development of ARDS, pneumonia, atelectasis, and pneumothorax, the cumulative use and duration of sedatives, and neuromuscular blocking agents, incidences of ICU delirium and ICU acquired weakness, patient–ventilator asynchrony and the need for decreasing of instrumental dead space.

3. STUDY DESIGN

PReVENT is an investigator–initiated, national, multicenter, parallel randomized controlled two–arm trial in intubated and ventilated intensive care unit patients without Acute Respiratory Distress Syndrome at start of ventilation. A cost–effectiveness analysis will be performed alongside the trial.

4. STUDY POPULATION

4.1 Population

We will recruit consecutive intensive care unit (ICU) patients who do not meet the consensus clinical definition for Acute Respiratory Distress Syndrome (ARDS) at onset of ventilation *and* who need invasive ventilation for an anticipated duration of > 24 hours. Patients are recruited in the ICUs of five centers in the Netherlands; three academic centers, and two teaching hospitals. Patients are screened for eligibility and randomized within an hour of initiation of invasive ventilation or, if already intubated and ventilated before admission, on admission to the ICU. We will randomize a total of 952 patients. It is expected that each participating center will randomize approximately 200 patients.

4.2 Inclusion criteria

In order to be eligible to participate in this trial, patients must meet all of the following criteria:

- Admission to an ICU participating in this trial
- Need for intubation and ventilation
- Within 1 hour of admission from the operation room or emergency room (if still intubated and ventilated), or within 1 hour of start of invasive ventilation in the ICU
- An expected duration of ventilation > 24 hours

4.3 Exclusion criteria

- Age less than 18 years
- Patients previously randomized in PReVENT
- Patients participating in other interventional trials
- Patients with a clinical diagnosis of ARDS or possible ARDS with a $\text{PaO}_2/\text{FiO}_2 < 200$ mm Hg (for reasons of safety, see insert below)
- Invasive ventilation longer than 12 hours directly preceding admission
- Patients with suspected or confirmed pregnancy
- Patients with increased and uncontrollable intracranial pressure (of ≥ 18 mmHg)
- Patients with GOLD classification III or IV chronic obstructive pulmonary disease (COPD)

- Patients with asthmatic status
- Patients with premorbid restrictive pulmonary disease (evidence of chronic interstitial infiltration on previous chest radiographs)
- Patients with new proven pulmonary thrombo–embolism
- Patients with any previous pneumectomy or lobectomy

The diagnosis of ARDS is clinical, requiring (a) a medical history, (b) the presence of bilateral opacities on the chest X–ray that are not fully explained by effusions, lobar/lung collapse, or nodules, and (c) respiratory failure not fully explained by cardiac failure or fluid overload. The $\text{PaO}_2/\text{FiO}_2$ is used to classify ARDS severity, with a $\text{PaO}_2/\text{FiO}_2$ between 200 and 300 mm Hg indicating mild ARDS, and a $\text{PaO}_2/\text{FiO}_2 < 200$ mm Hg indicating moderate or severe ARDS.

This diagnostic approach, however, could be difficult if not impossible in ICU patients within the first hour after intubation and start of ventilation: they frequently suffer from temporary post–intubation atelectasis as a reason for a low $\text{PaO}_2/\text{FiO}_2$, the medical history is often not yet complete, and imaging studies are usually not yet performed or the results available. The risk is that only the $\text{PaO}_2/\text{FiO}_2$ is used to diagnose ARDS in the short time frame after intubation, which could induce severe bias, as many of these patients do not have ARDS.

Thus, we exclude all patients that are clinically diagnosed with ARDS. Patients with a $\text{PaO}_2/\text{FiO}_2 < 200$ mm Hg are also excluded since we consider these patients at high risk of having ARDS; only when the attending physician explicitly states the patient has no ARDS *and* no direct risk factor for ARDS is present, the patient can be included. Patients without ARDS and with a $\text{PaO}_2/\text{FiO}_2$ between 200 and 300 mm Hg can be included: as these patients very seldom have ARDS.

4.4 Sample size

The required sample size is calculated using data from the recently published meta–analysis [12] and a secondary analysis of this meta–analysis using individual patient data from the studies performed in ICU patients [27]. The sample size is computed on the basis of the hypothesis that ventilation with

lower tidal volumes is associated with a reduction of one day of ventilation. A sample size of 397 patients in each group has 80% statistical power to detect a difference of one ventilator-free day and alive at day 28 after ICU admission, with means of 23 and 24 days respectively. Assuming that the common standard deviation is 5 using a two group t-test with a 0.05 two-sided significance level. The sample size is increased by 20% to correct for dropouts and lost to follow up meaning that each group will contain 476 patients.

5. TREATMENT OF SUBJECTS

5.1 Treatment

Patients randomized to the 'lower tidal volume'–arm are ventilated with a tidal volume of 4 to 6 ml/kg predicted body weight (PBW); patients randomized to the 'higher tidal volume'–arm are ventilated with a tidal volume of 8 to 10 ml/kg PBW. Patients are randomly assigned in a 1:1 ratio to the 'lower tidal volume'–arm or the 'higher tidal volume'–arm.

5.2 Ventilation protocol

The allowed ventilation modes are volume controlled ventilation and pressure support ventilation (i.e., pressure controlled ventilation, combined modes volume controlled and pressure controlled ventilation, and automated modes of ventilation are NOT allowed). The inspiration–to–expiration ratio with volume controlled ventilation is 1:2. With volume controlled ventilation the inspiration time and pause are set at 25% and 10% respectively. With pressure support ventilation the highest possible pressure rise is chosen, and cycling off is set at 25%.

The inspired oxygen fraction is 0.21 or higher to maintain oxygen saturation 90 to 92% and/or $\text{PaO}_2 > 7.3$ to 10.7 kPa (55 to 80 mmHg). The respiratory rate is adjusted to maintain a blood pH of 7.25 to 7.45. In case of metabolic acidosis or – alkalosis, a lower or higher than normal PaCO_2 can be accepted, left to the discretion of the attending physician. The lowest level of positive end–expiratory pressure is 5 cmH₂O; allowed FiO_2 –PEEP–combinations are provided in table 1. Recruitment maneuvers are allowed, when deemed necessary, left to the discretion of the attending physician.

In both arms a tidal volume is titrated per PBW, which is calculated according to a previously used formula [6]: $50 + 0.91 \times (\text{centimeters of height} - 152.4)$ for males and $45.5 + 0.91 \times (\text{centimeters of height} - 152.4)$ for females.

5.2.1 Lower tidal volume ventilation

Patients randomized to the 'lower tidal volume'–arm start with a tidal volume of 6 ml/kg PBW. The tidal volume size is decreased in steps of 1 ml/kg PBW per hour, to a minimum of 4 ml/kg PBW, unless the patient suffers from severe dyspnea (identified by increased respiratory rate > 35 breaths per minute accompanied by increasing levels of discomfort with or without need

for more sedation) or unacceptable acidosis (table 1). The following measures can be taken to prevent

Table 1. Ventilator settings with the 2 ventilation strategies		
	Lower tidal volume ventilation	Higher tidal volume ventilation
Ventilator mode	Assisted MV= volume controlled Spontaneous MV = pressure support	Assisted MV= volume controlled Spontaneous MV = pressure support
Target tidal volume	4 ml/kg PBW	10 ml/kg PBW
Allowable tidal volume in case of high pressures (in the conventional group) or severe dyspnea (in the protective ventilation group)	6 ml/kg PBW*	8 ml/kg PBW*
Allowable ventilator rate setting needed to achieve normal pH (7.25 to 7.45)	6 to 35 breaths/min; typically higher than with an higher tidal volume	6 to 35 breaths/min; typically lower than with a lower tidal volume
Inspiration-to-expiration ratio	1:2	1:2
PaO ₂ or SpO ₂ -targets	7.3 to 10.7 kPa (55 to 80 mmHg), or 90 to 92%	7.3 to 10.7 kPa (55 to 80 mmHg), or 90 to 92%
Allowable combinations of FiO ₂ and PEEP	0.21 and 5 cm H ₂ O 0.30 and 5 cm H ₂ O 0.40 and 5 cm H ₂ O 0.40 and 8 cm H ₂ O 0.50 and 8 cm H ₂ O 0.50 and 10 cm H ₂ O 0.60 and 10 cm H ₂ O	0.21 and 5 cm H ₂ O 0.30 and 5 cm H ₂ O 0.40 and 5 cm H ₂ O 0.40 and 8 cm H ₂ O 0.50 and 8 cm H ₂ O 0.50 and 10 cm H ₂ O 0.60 and 10 cm H ₂ O
Daily assessment of the ability to breathe with pressure support	Required when FiO ₂ ≤ 0.4 or earlier	Required when FiO ₂ ≤ 0.4 or earlier
Allowable tidal volume when ARDS develops	6 ml/kg PBW	6 ml/kg PBW
Allowable tidal volume when ARDS develops; in case of severe dyspnea or unacceptable pH	8 ml/kg PBW	8 ml/kg PBW
Allowable combinations of FiO ₂ and PEEP (continued)	0.21 and 5 cm H ₂ O 0.30 and 5 cm H ₂ O 0.40 and 5 cm H ₂ O 0.40 and 8 cm H ₂ O 0.50 and 8 cm H ₂ O 0.50 and 10 cm H ₂ O 0.60 and 10 cm H ₂ O 0.70 and 10 cm H ₂ O 0.70 and 12 cm H ₂ O 0.70 and 14 cm H ₂ O 0.80 and 14 cm H ₂ O 0.90 and 14 cm H ₂ O 0.90 and 16 cm H ₂ O 0.90 and 18 cm H ₂ O 1 and 18 cm H ₂ O 1 and 20 cm H ₂ O 1 and 22 cm H ₂ O 1 and 24 cm H ₂ O	0.21 and 5 cm H ₂ O 0.30 and 5 cm H ₂ O 0.40 and 5 cm H ₂ O 0.40 and 8 cm H ₂ O 0.50 and 8 cm H ₂ O 0.50 and 10 cm H ₂ O 0.60 and 10 cm H ₂ O 0.70 and 10 cm H ₂ O 0.70 and 12 cm H ₂ O 0.70 and 14 cm H ₂ O 0.80 and 14 cm H ₂ O 0.90 and 14 cm H ₂ O 0.90 and 16 cm H ₂ O 0.90 and 18 cm H ₂ O 1 and 18 cm H ₂ O 1 and 20 cm H ₂ O 1 and 22 cm H ₂ O 1 and 24 cm H ₂ O
Abbreviations: MV, mechanical ventilation; PBW, predicted body weight; PEEP, positive end-expiratory pressure; FiO ₂ , O ₂ -fraction of inspired air; ARDS, acute respiratory distress syndrome. *, with pressure support tidal volumes could be higher than targeted with the lowest possible pressure support level, or lower than targeted with the highest allowed maximum airway pressure - this is accepted and never a reason for use of (more) sedation, paralytic agents, and/or switch to a controlled mode of ventilation.		

respiratory acidosis: increasing respiratory rate and decreasing instrumental dead space by shortening ventilation tubing, to limit dead space ventilation.

Patients randomized to the lower tidal volume arm may need very little support when the ventilator is switched to pressure support ventilation, but a minimum of 5 cm H₂O should be used. In case the resulting tidal volume exceeds 6 ml/kg PBW this must be accepted (i.e., this is neither a reason to use (more) sedation and/or muscle relaxants, nor to switch to volume controlled ventilation) (see weaning).

5.2.2 Higher tidal volume ventilation

Patients randomized to the 'higher tidal volume'–arm start with a tidal volume of 10 ml/kg PBW. With volume–controlled ventilation the plateau pressure should not exceed 25 cm H₂O [2]. Only if the plateau pressure exceeds 25 cm H₂O the tidal volume is decreased in steps of 1 ml/kg PBW per hour, to a minimum of 8 ml/kg PBW (table 1). With pressure support, tidal volume titration is by variation of the pressure support level. Other modes of ventilation are *not* allowed.

Patients randomized to the higher tidal volume arm generally need more support when the ventilator is switched to pressure support ventilation, but the maximal airway pressure should not exceed 25 cm H₂O [2]. In case the resulting tidal volume remains below 10 ml/kg PBW this must be accepted (i.e., this is neither a reason to use (more) sedation and/or muscle relaxants, nor to switch the ventilator to volume–controlled ventilation) (see weaning). Other modes of ventilation are *not* allowed.

5.2.3 Ventilation for patients who develop ARDS

When a patient develops ARDS, ventilation should be continued according to the existing guidelines, which means that the tidal volume must be reduced to 6 ml/kg PBW. Allowed tidal volumes and FiO₂–PEEP–combinations are provided in table 1.

5.2.4 Weaning

Daily assessment of the ability to breathe with pressure support ventilation is required as soon as FiO₂ ≤ 0.4 or when the PEEP level and FiO₂ level are lower than the day before. Other modes of ventilation are *not* allowed.

In addition, the ventilator can be switched to pressure support ventilation at any moment the attending nurse or physician considers the

patient is awake enough to breathe with pressure support ventilation. Assessment of the ability to breathe with pressure support is also required in case patient–ventilator asynchrony is noticed (ineffective breathing; double triggering, use of accessory respiratory muscles). A patient is assumed to be ready for extubation when the following criteria are met for at least 30 minutes; the attending physician makes the final decision for extubation:

- Responsive and cooperative
- Adequate cough reflex
- $\text{PaO}_2/\text{FiO}_2$ of > 200 mmHg with $\text{FiO}_2 \leq 40\%$
- Respiratory rate of 8 to 30/minute
- No signs of respiratory distress (i.e., marked accessory muscle use, abdominal paradox, diaphoresis, marked dyspnea)
- Pressure support level < 7 cm H₂O (lower tidal volume arm) or < 12 cm H₂O (higher tidal volume arm)
- Hemodynamically stable (systolic blood pressure 80 to 160 mmHg and heart rate 40 to 130/min) and no uncontrolled arrhythmia
- Temperature $> 36.0^\circ\text{C}$ and $< 38.5^\circ\text{C}$

In the higher tidal volume arm physicians and nurses may decide to lower the pressure support level first (i.e., before extubation). For this, the pressure support level is lowered step–wise with steps of 2 to 5 cm H₂O per hour to < 7 cm H₂O. If this is not tolerated according to the conditions mentioned above, the pressure support level is set back to maintain a tidal volume as per randomization and the patient is assessed for extubation the following day.

If a patient becomes able to breathe without assistance but subsequently requires additional ventilation within 28 days after randomization, the same tidal volume protocol is resumed. Non–invasive ventilation is allowed, but it should be tried to have comparable tidal volumes as with invasive ventilation, as per randomization.

5.2.5 Tracheostomy

Early tracheostomy has no advantage over late tracheotomy [28]. Therefore, tracheostomy is only to be performed on strict indications and preferably not earlier than 10 days after intubation.

Strict indications for tracheostomy:

- Failure to intubate
- Expected duration of ventilation > 14 days
- Glasgow Coma Score < 7 and/or inadequate swallow or cough reflex with retention of sputum
- Severe ICU–acquired weakness
- Prolonged or unsuccessful weaning
- Repeated respiratory failure after extubation

Weaning with a tracheostomy follows the rules as described above under ‘weaning’, a suggested scheme for unassisted ventilation with a tracheostomy is described in APPENDIX II.

5.3 Sedation protocol

Sedation follows the local guidelines for sedation in each participating units. In general, these guidelines favor the use of analgo–sedation over hypno–sedation, use of bolus over continuous infusion of sedating agents, and the use of sedation scores.

Nurses determine the level of sedation at least 3 times per day. The adequacy of sedation in each patient is evaluated using a Richmond Agitation Sedation Scale (RASS) [29, 30]. A RASS score of –2 to 0 is seen as adequate sedation. As stated above, sedation adjustments should never be done to allow a lower or higher tidal volume. The goals of sedation are to reduce agitation, stress and fear; to reduce oxygen consumption (heart rate, blood pressure and minute volume are measured continuously); and to reduce physical resistance to– and fear of daily care and medical examination. Patient comfort is the primary goal.

Level of pain is determined using scales such as Numeric Rating Scale (NRS), Visual Analogue Scale (VAS), Critical Care Pain Observation Tool (CCPOT) or Behavioral Pain Scale (BPS).

5.4 Other ICU protocols

5.4.1 Selective oropharyngeal– or digestive tract decontamination

To prevent nosocomial infections, selective oropharyngeal decontamination (SOD) or selective decontamination of the digestive tract (SDD) is performed in all patients who are expected to need ventilation for longer than 48 hours, and/or are expected to stay in ICU for longer than 72 hours [31].

5.4.2 Thrombosis prophylaxis

Thrombosis prophylaxis is indicated for all patients who are not treated with anticoagulants, e.g. for therapeutic reasons or systemic prophylaxis because of an implanted device or extra–corporal circulation like veno–venous hemofiltration. Thrombosis prophylaxis will be given according to local guidelines.

5.4.3 Fluid regimens

A fluid balance targeted at normovolemia and a diuresis of ≥ 0.5 ml/kg/hour should be maintained. Crystalloid infusions are preferred over colloid infusions.

5.4.4 Nutrition

A hypo–caloric, protein–rich diet (1.2–1.7 gr/kg bodyweight /24 hours) is started as soon as possible after ICU admission. Enteral nutrition with a feeding gastric tube is the preferred over intravenous feeding. If stomach retention occurs a duodenal tube can be used if administration of prokinetics is not sufficient, according to local guidelines. When optimal protein intake cannot be reached within 4 days, additional parenteral nutrition (TPV) can be started.

6. METHODS

6.1 Study procedures

Patients in participating intensive care units (ICU) are screened and randomized within 1 hour of start of mechanical ventilation in the unit. Demographic data on screened patients regardless of meeting enrollment criteria will be recorded (registry: age, gender, type of surgery).

6.2 Randomization, blinding and treatment allocation

Randomization will be performed using a dedicated, password protected, SSL–encrypted website. Randomization sequence is generated by a dedicated computer randomization software program using random block sizes and is stratified per center and per intubation location (i.e., in the ICU or before ICU admittance in the operation room or in the emergency room). No blocking is applied to other trial factors. Due to the nature of the treatment, blinding is not possible.

Patients are randomly assigned in a 1:1 ratio to lower tidal volume ventilation (4 to 6 ml/kg PBW) (the ‘lower tidal volume’–arm) or ventilation with higher tidal volumes (8 to 10 ml/kg PBW) (the ‘higher tidal volume’–arm).

6.3 Study parameters/endpoints

6.3.1 Main study parameter

The primary endpoint is the number of ventilator–free days and alive at day 28, defined as the number of days from day 1 to day 28; the patient is alive and breathes without assistance of the mechanical ventilator, if the period of unassisted breathing lasted at least 24 consecutive hours.

6.3.2 Secondary study parameters

Secondary study parameters include:

- ICU length of stay (LOS)
- Hospital LOS
- ICU mortality
- Hospital mortality
- 90–day mortality
- Development of ARDS (APPENDIX I)
- Ventilator–associated pneumonia (APPENDIX I)
- Atelectasis, if a chest X–ray is obtained (APPENDIX I)

- Pneumothorax, if a chest X-ray or other kind of imaging suitable for diagnosing pneumothorax is obtained (APPENDIX I)
- Cumulative use and duration of use of sedatives
- Cumulative use and duration of use of neuromuscular blocking agents (other than used for intubation)
- ICU–delirium (see below)
- ICU–acquired weakness (see below)
- Transfusion of blood products, including packed red blood cells, plasma and platelet concentrates
- Need for decreasing dead space
- Changes in EAdi–signals (in selected patients, see below)
- Volatile organic compound composition of exhaled air: the exhaled breath profile obtained by electronic nose, abundance of VOCs in the exhaled breath assessed by gas–chromatography and mass–spectrometry and selective ion flow tube mass–spectrometry (in selected centers, see below)

6.3.3 Other study parameters

Health care costs will be estimated from a health systems perspective and include costs of ventilation, costs of stay in ICU, costs of stay in hospital, costs of cumulative use of sedatives, costs of neuromuscular blocking agents, costs of use of tracheostomies, costs of ventilator–associated pneumonia.

These are used to calculate:

- Incremental cost per ICU–day avoided
- Incremental cost per mechanical ventilation–day avoided

6.4 Data collection

- On admission and within first 24 hours:
 - Gender and age (male + years)
 - Height and weight (kg + cm)
 - Reason for ICU admission
 - Reason for ventilation
 - Cause of respiratory failure
 - APACHE II score and SAPS II score
- Respiratory status, on admission, and every day until day 28:

- Intubation status (if extubated: time of extubation)
- Tracheostomy status (if tracheostomized: time of tracheostomy)
- Invasiveness of ventilation (invasive, non–invasive, or intermittent ventilation via tracheostomy)
- Need for decreasing instrumental dead space
- Location of patient, every day until day 28, and at day 90 (in ICU, hospital, other facility, or home) and life status (alive or deceased)
- Pulmonary complications, every day until day 28 or discharge from ICU, whatever comes first:
 - Moderate or severe ARDS (yes or no) (APPENDIX I)
 - Pneumonia (yes or no) (APPENDIX I)
 - Atelectasis (yes or no) (APPENDIX I)
 - Pneumothorax (yes or no) (APPENDIX I)
- Sedation and neuromuscular blocking agents, every day until day 28 or discharge from ICU, whatever comes first:
 - Cumulative use and duration of sedatives (yes or no, and cumulative dose, mg)
 - Neuromuscular blocking agents (yes or no, and cumulative dose, mg)
 - Richmond Agitation Sedation Scale (RASS) score [29, 30]. All patients with a RASS score of ≥ -2 should be tested for delirium using the CAM–ICU (see below).
- ICU delirium, every day until day 28 or discharge from ICU, whatever comes first:
 - Confusion Assessment Method for ICU (CAM–ICU) score [32, 33].
- ICU–acquired weakness, every day until day 28 or discharge from ICU, whatever comes first:
 - Medical Research Council (MRC) score [34] (APPENDIX I)
 - Strength assessment (only in selected centers: Grip strength assessment [35])

6.4.1 Other data to be collected

- Mechanical ventilation parameters, 1 hour before and 1 hour after randomization, and every day at a fixed time point until cessation of ventilation:

- Tidal volume (ml + ml/kg PBW)
- Respiratory rate (breath/minute)
- Level of positive end–expiratory pressure (cmH₂O)
- Peak and plateau pressures, or level of pressure support (level above PEEP, and maximal airway pressure, cmH₂O)
- Inspiration to expiration ratio
- Inspired oxygen fraction (%)
- Minute volume (liters/minute)
- Pulmonary compliance (ml/ cmH₂O)
- Lung Injury Score [36]
- Oxygenation Index [37]
- Respiratory parameters, 1 hour before and 1 hour after randomization, and every day at a fixed time point until cessation of ventilation:
 - Peripheral oxygen saturation (%)
 - End–tidal fractions of CO₂ (kPa)
 - PaO₂ (kPa)
 - PaCO₂ (kPa)
 - Arterial bicarbonate (mmol/L)
 - Arterial pH
 - Arterial base excess (mmol/L)
- Non–respiratory parameters, every day until day 28 or discharge from ICU, whatever comes first:
 - Transfusions of blood products (type and ml)
 - Infusion of crystalloids (type and ml)
 - Infusion of (artificial) colloids (type and ml)
 - Cumulative fluid balance (ml)
 - Cumulative urine output (ml)
- Sequential Organ Failure Assessment score (SOFA) score [38]
- Other data:
 - Diaphragm weakness development; assessed by measuring the electrical activity of the diaphragm (EAdi) using a ‘Neurally Adjusted Ventilatory Assist’ (NAVA)–catheter. After placement of a NAVA catheter data are available continuously. This is only done in patients

connected to a ventilator with hardware for NAVA ventilation, *and* a NAVA catheter in use (see APPENDIX III)

- Exhaled air collection; continuous analysis of exhaled breath with sensor technology (electronic nose analysis) and selective ion flow tube mass-spectrometry and twice daily collection of exhaled breath onto a sorbent tube for gas-chromatography and mass-spectrometry. This is only done in patients ventilated in centers that can collect air samples for breath-analysis (see APPENDIX IV)

6.5 Withdrawal of individual subjects

Subjects can leave the study at any time for any reason if they wish to do so without any consequences. The investigator can decide to withdraw a subject from the study for urgent medical reasons.

6.6 Replacement of individual subjects after withdrawal

When deferred consent is not obtained after randomization and provisional inclusion of a patient, replacement of the randomization subject will be done. In the randomization log these cases will be recorded without patient-specific data. The randomization subject will be replaced in order to retain properly distributed randomization groups and stratification.

6.7 Follow-up of subjects withdrawn from treatment

Patients will not be subjected to follow-up.

7. SAFETY REPORTING

7.1 Section 10 WMO event

In accordance to section 10, subsection 1, of the WMO, the investigator will inform the subjects and the reviewing accredited METC if anything occurs, on the basis of which it appears that the disadvantages of participation may be significantly greater than was foreseen in the research proposal. The study will be suspended pending further review by the accredited METC, except insofar as suspension would jeopardize the subjects' health. The investigator will take care that all subjects are kept informed.

7.2 SAEs

This study compares two treatment strategies that are used in standard care, therefore no related serious adverse events (SAEs) are expected.

7.3 Data Safety Monitoring Board (DSMB)

The DSMB will be composed of 5 individuals, one of which will be the chairman.

- The DSMB will first meet approximately 6-8 months after the first patient is enrolled; the first meeting will be scheduled after the first 150 patients
- Subsequent to this meeting the DSMB will meet virtually every 6 months
- All unexpected adverse events will be reported to the DSMB
- The DSMB will review the overall status of the program: number of patients enrolled overall and in each center, adherence to the protocol overall and by each center
- The DSMB may be composed of the following individuals:
 - Prof. dr. Antonio Artigas, Hospital de Sabadell, Sabadell, Spain
 - Prof. dr. Thomas Bein, Regensburg University Hospital, Regensburg, Germany
 - Prof. dr. Ognjen Gajic, Mayo Clinics, Rochester, MN
 - Dr. Diederik Gommers, Erasmus MC, Rotterdam, The Netherlands
 - Prof. dr. Herman Wrigge, University of Leipzig Hospital, Leipzig, Germany

The report and/ or advice(s) of the DSMB will only be sent to the sponsor of the study, the Academic Medical Center. Should the sponsor decide not to fully implement the advice of the DSMB, the sponsor will send the advice to

the reviewing METC, including a note to substantiate why (part of) the advice of the DSMB will not be followed.

8. STATISTICAL ANALYSIS

8.1 Primary study parameter(s)

The primary outcome, the number of ventilator-free days and alive at day 28 after ICU admission, is analyzed using Cox's regression. Possible imbalance between groups will be modeled in the Cox model. P-values of 0.05 are used for statistical significance. When appropriate, statistical uncertainty will be expressed by the 95% confidence levels. All statistical analyses will be performed with the R language and environment for statistical computing.

8.2 Secondary study parameter(s)

Continuous normally distributed variables will be expressed by their mean and standard deviation or when not normally distributed as medians and their interquartile ranges. Categorical variables will be expressed as n (%). To test groups Student's t test will be used, if continuous data is not normally distributed the Mann-Whitney U test will be used. Categorical variables will be compared with the Chi-square test or Fisher's exact tests. Time dependent data will be analyzed using a proportional hazard model adjusted for possible imbalances of patients' baseline characteristics. Analysis will be performed with R statistics version 3.0.2. Patient characteristics will be compared and described by appropriate statistics.

8.3 Primary and secondary analyses

The goal of the primary analysis is to quantify the effect of lower tidal volumes vs. higher tidal volumes on the number of ventilator free days and alive at day 28. Statistical analysis will be based on the intention-to-treat principle. We will also perform a per-protocol analysis, comparing patients who received lower tidal volumes and patients who received higher tidal volumes. Other secondary analyses include analysis of patients who had pneumonia versus patients without pneumonia,, and patients with sepsis versus patients without sepsis.

COST-EFFECTIVENESS ANALYSIS

8.4 Cost-effectiveness endpoints

The primary cost-effective (CE) measure will be incremental cost per ICU day avoided; the secondary CE measure is incremental cost per mechanical ventilation day avoided.

8.5 Perspective on costs and time horizon of the analysis

Costs will be considered from a health systems perspective over the time horizon of the analysis. As this time horizon is shorter than 1 year, discounting of future costs and effects is not applicable.

8.6 Collection of resource use and cost data

Collection and analysis of cost data is fully integrated into the study. Analyses of effect parameters are guided by the above described analysis plan and hypotheses. Also in accordance with the guideline, analyses are conducted with an intention-to-treat approach.

Per patient volumes of resource use associated with mechanical ventilation, stay in ICU, stay in hospital (other wards), and cumulative use of sedatives, neuromuscular blocking agents, tracheostomies, and ventilator-associated pneumonia, will be collected from the hospitals' administration databases. Unit prices for each cost parameter will be established using the hospitals' financial data, where available, or national tariffs as recommended in the Costing Guideline published by the Dutch Health Insurance Council (CVZ) [39]. Costs are calculated as volume * unit price.

8.7 Analysis and presentation of Incremental Cost-effectiveness

Ratios

Incremental Cost-Effectiveness Ratios (ICER) will be calculated by dividing the difference in costs between the two ventilation strategies by the difference in effect between the two ventilation strategies. Cost data do not conform to the assumptions for standard statistical tests for comparing differences in arithmetic means, i.e. they are usually right-skewed and always truncated at zero, Therefore the nonparametric bootstrap is the appropriate method to compare means and calculate 95% confidence intervals around the ICER, and analyse the probability of the ventilation strategy with low tidal volume being cost-effective. Bootstrapping is performed using Excel® software

(Microsoft, Redmond, WA) Results are presented on the cost–effectiveness graph, plotting the incremental costs against the incremental effectiveness.

8.8 Sensitivity Analyses

One and multi-way sensitivity analyses will be applied to test the robustness of the ICER against parameter uncertainty, by varying the parameter inputs over a predefined range of alternative inputs. Probabilistic sensitivity analysis (PSA), using 10,000 second order Monte-Carlo simulations, will be employed to handle uncertainty in the cost-effectiveness analysis by obtaining distributions of costs, effects and the resulting cost-effectiveness estimates. For this, distributions will be fitted to the effect and cost parameters using the method of moments approach, i.e. equating the mean and SE observed in the trial data to the expressions for the mean and SE of the appropriate distribution. The parameters of the distribution (α , β) can then be solved analytically. Gamma distributions will be fitted to costs parameters. Normal or lognormal distributions will be fitted to the effect parameters, depending on the observed distribution of these data. Sensitivity analysis will be performed in combined Excel® and R software.

8.9 Subgroup analyses

As we expect variation in cost structures between academic and non–academic ICUs and hospitals, all analyses will also be performed for these subgroups using mean cost data specific to academic and non–academic centres.

8.10 Analysis of missing data

In case of >10% missing data, these will be imputed using the SPSS (IBM, Armonk, NY) multiple imputation algorithm, to reflect the uncertainty that is inherent when replacing missing data.

9. ETHICAL CONSIDERATIONS

9.1 Regulation statement

This study will be conducted according to the principles of the Declaration of Helsinki as stated in the current version of Fortaleza, Brazil, 2013 and in accordance with the Medical Research Involving Human Subjects Act (WMO).

9.2 Recruitment and consent

9.2.1 Deferred consent

For this study we ask for deferred consent and we appeal to the emergency procedure for consent in medical research as stated in article 6, paragraph 4 of the WMO because of the following reasons.

Mechanical ventilation is one of the most often applied therapies in the ICU. Patients who are in respiratory distress are in an emergency situation and need immediate ventilator support, start of ventilation cannot be postponed. Patients admitted for ventilation to an Intensive Care Unit (ICU) are, without exception, incompetent to give informed consent and obtaining informed consent from a legal representative takes on average up to half a day, even by an experienced research team.

We propose to randomize each patient who meets the inclusion criteria ultimately within 1 hour after ICU admission (if ventilation started before admission), or within 1 hour after intubation and start of ventilation (if ventilation started after admission). Ventilation according to randomization is started immediately thenceforth. Informed consent from the legal representative will be requested as soon as possible thereafter, but never later than 24 hours after randomization. If informed consent is not obtained within those 24 hours, or if a legal representative denies participation within this time frame, the patient is excluded and data will no longer be used. Thenceforth the patient is ventilated according to the policy of the attending physician.

9.2.2 Injurious effects of ventilation

Ventilation can harm the lungs and respiratory muscles, even after a short period of ventilation. If a patient, in the proposed trial, is already ventilated for several hours, effects of ventilation could already be in place, largely reducing validity of the trial outcomes.

From experimental animal studies we know that mechanical ventilation

can cause effects within an hour of ventilation with a higher tidal volume [40]. These findings are in line with results from clinical studies, showing ventilator-related effects after relative short periods of ventilation, e.g. after ventilation during general anesthesia for surgery [41]. Two recently published randomized controlled trials of patients undergoing major surgery, comparing a ventilation strategy including use of a lower tidal volume with a ventilation strategy including use of a higher tidal volume, showed an important effect of ventilation on the incidence of postoperative pulmonary complications [42, 43].

9.2.3 Time between asking for and obtaining informed consent

Most critically ill patients who need ventilation cannot be approached for informed consent for a study at ICU admission. Indeed, those patients are usually in severe respiratory distress, sedated or in coma. A prospective observational study on study recruitment practices in critically ill patients performed by a respected and experienced research group in Canada [44] showed that time from recognizing study eligibility to obtaining informed consent by a legal representative was as high as 12 hours, even while time from recognition to the first contact with a legal representative was as short as 2 hours. This is in line with our own experience. In a recent observational study in the Academic Medical Center, Amsterdam, the Netherlands, time from recognizing of study eligibility to obtaining informed consent from a legal representative was as high as 14 hours (study ID NL34294.018.10). These findings are quite understandable if we realize that relatives of a severely ill patient do not oversee the situation in the circumstances of emergency, rendering them incapable of making a balanced decision [44–46].

The experience of ICU patients enrolled under deferred consent is mainly positive. To investigate contentment of patients that were included using deferred consent, a questionnaire was designed for– and distributed under the participants of the large NICE–SUGAR trial [47], a trial that compared a strict blood glucose control strategy with one that accepts higher blood glucose levels [48]. Of the responders (79% of all participants), a large majority (96%) said to have granted consent if they would have been asked. A large majority (93%) mentioned they were happy with the decision made by the representative at the moment they were incapable of giving informed

consent [48].

9.2.4 Ethical aspects

We can underpin the idea of 'clinical equipoise' [49]. Both ventilation strategies are used interchangeably in ICUs worldwide and are considered safe. We do not know what the best ventilation strategy is. The most recent observational study of ventilation practice in ICU patients shows that the median tidal volume size is 7.5 [6.1 to 8.7] ml/kg predicted body weight (PBW) [20]. This is in line with a recent meta-analysis of four observational studies of patients without the Acute Respiratory Distress Syndrome (ARDS) in the United States, Finland and the Netherlands between 2007- 2013 [22–25], that shows the used tidal volume size is 8 ml/kg PBW [yet unpublished data]. This meta-analysis also showed there is large variability in tidal volume size, as illustrated in figure 1. Notably, 11% of patients is ventilated with a tidal volume of 4 to 6 ml/kg PBW, and 53% is ventilated with a tidal volume of 8 to 10 ml/kg PBW; the two strategies in the proposed trial.

9.2.5 Similarities with previous studies using deferred consent

The proposed trial has many similarities with recently published interventional trials in ICU patients using deferred consent [47, 50–53]. In a French trial in ICU patients with hypovolemic shock who needed immediate resuscitation with intravenous fluids, colloid infusion was compared with crystalloid infusion [50]. An international ICU study compared resuscitation with 4% albumin infusion with normal saline infusion [51]. In the RENAL replacement therapy study in critically ill patients with acute kidney injury, continuous venovenous hemodiafiltration with either a lower or higher intensity was compared [52]. One Dutch trial in ICU patients with hypovolemic shock compared lactate-guided fluid resuscitation with standard resuscitation [53]. In these trials, alike in the proposed trial, patients were incapable at the moment therapy had to be started, the effects of the interventions were for a major part depending on its start, and the strategies under study were both used in daily practice.

9.2.6 Conclusions

Critically ill patients in need of ventilation are, without exception, incapable at the moment of ICU admission. Obtaining informed consent from a legal representative takes many hours. Start of ventilation in patients who need ventilatory support cannot wait. Both ventilation strategies are used

interchangeably in ICUs worldwide and are considered safe.

9.3 Benefits and risks assessment, group relatedness

Burden and risks of the two ventilation strategies are uncertain. Lower tidal volume ventilation could induce an increased need for sedation and increase the risk of asynchrony and respiratory muscle fatigue, while ventilation with higher tidal volumes could increase the risk of lung injury. Both strategies are currently used; there is no additional risk for patients enrolled in this study compared to standard care.

We specifically chose *not* to exclude incompetent patients for two reasons. First, most if not all severely ill patients needing mechanical ventilation should be considered incompetent due to their needs for continuous sedation. Second, the strategies to be compared in this study are to be used in severely ill, intubated and ventilated patients. These conditions are not present in patients who are not suffering from a critical disease. We therefore consider it impossible not to include these patients in a study comparing strategies for mechanical ventilation to be used in that specific patient group.

9.4 Compensation for injury

The sponsor/investigator has a liability insurance, which is in accordance with article 7 subsection 6 of the WMO. As this study compares two treatments used for standard care an exemption from the requirement to insure cover for damage to research subjects through injury or death caused by the study is applicable.

10.ADMINISTRATIVE ASPECTS, MONITORING AND PUBLICATION

10.1 Handling and storage of data and documents

All patients will be addressed to with a random patient identification code. The codebook will be stored digitally and in paper. The paper version will be stored behind a lock and the digital form will be encrypted with a double password. All data will be stored for the length of the study and for 15 years afterwards, for further publication. All handling of personal data will comply with the Dutch Personal Data Protection Act.

10.2 Monitoring and Quality Assurance

Queries on the database will be done by a statistician and analyzed by the monitor to signalize early aberrant patterns, trends, issues with consistency or credibility and other anomalies. On-site monitoring will comprise controlling presence and completeness of the research dossier and the informed consent forms, source data checks will be performed as described in the monitoring plan. Every participating center will be visited at least once every year. The monitor of this study is: A.L.I.P. van der Veen (Academic Medical Center, University of Amsterdam, Amsterdam, The Netherlands). The details of monitoring are described in the monitoring plan.

10.3 Amendments

Amendments are changes made to the research after a favorable opinion by the accredited METC has been given. All amendments will be notified to the METC that gave a favorable opinion. All substantial amendments will be notified to the METC and to the competent authority. Non-substantial amendments will not be notified to the accredited METC and the competent authority, but will be recorded and filed by the sponsor.

10.4 Annual progress report

The sponsor/investigator will submit a summary of the progress of the trial to the accredited METC once a year. Information will be provided on the date of inclusion of the first subject, numbers of subjects included and numbers of subjects that have completed the trial, unexpected problems and amendments.

10.5 End of study report

The investigator will notify the accredited METC of the end of the study within a period of 8 weeks. The end of the study is defined as the 90th day after the

last patients' admittance to the ICU. In case the study is ended prematurely, the investigator will notify the accredited METC within 15 days, including the reasons for the premature termination. Within one year after the end of the study, the investigator/sponsor will submit a final study report with the results of the study, including any publications/abstracts of the study, to the accredited METC.

10.6 Public disclosure and publication policy

The study protocol and analysis plan will be published before start of the study on Clinicaltrials.gov. The results of the study will find their way into (inter-) national scientific journals and guidelines. We will submit analyses to scientific journals in the field of intensive care medicine as well as anesthesiology, since both ICU physicians and anesthesiologists apply ventilation in the ICU setting.

11.STRUCTURED RISK ANALYSIS

11.1 Synthesis

To remark anomalies in an early stage, statistical queries on the collected data will be done by a statistician and judged by the monitor as described in the monitoring plan (see monitoring plan). An independent Data Safety Management Board (DSMB) is installed to monitor safety parameters and the overall conduct of the study (see above).

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APPENDIX I

Table 1. The Berlin definition for ARDS [54]

Timing	Within 1 week of a known clinical insult , or new/worsening respiratory symptoms		
Chest imaging ^a	Bilateral opacities —not fully explained by effusions, lobar/lung collapse, or nodules		
Origin of edema	Respiratory failure not fully explained by cardiac failure or fluid overload; need objective assessment (e.g., echocardiography) to exclude hydrostatic edema if no risk factor present		
Oxygenation ^b	Mild 200 < PaO ₂ /FiO ₂ ≤ 300 mmHg 26.7 < PaO ₂ /FiO ₂ ≤ 40 kPa with PEEP ≥ 5 cmH ₂ O	Moderate 100 < PaO ₂ /FiO ₂ ≤ 200 mmHg 13.3 < PaO ₂ /FiO ₂ ≤ 26.7 kPa with PEEP ≥ 5 cmH ₂ O	Severe PaO ₂ /FiO ₂ ≤ 100 mmHg PaO ₂ /FiO ₂ ≤ 13.3 kPa with PEEP ≥ 5 cmH ₂ O

^aChest X-ray or CT scan; ^bIf altitude higher than 1000 m, correction factor should be made as follows: PaO₂/FiO₂ 9 (barometric pressure/760); ^cThis may be delivered non-invasively in the mild ARDS group. Abbreviations: ARDS, acute respiratory distress syndrome; PaO₂, partial pressure of arterial oxygen; FiO₂, fraction of inspired oxygen; PEEP, positive end-expiratory pressure; CPAP, continuous positive airway.

DEFINITIONS

- **Pneumonia:** new or progressive radiographic infiltrate plus at least two of the following: fever tympanic temperature >38,5, leucocytosis or leucopenia and/or purulent secretions
- **Atelectasis:** lung opacification on chest radiograph with shift of the mediastinum, hilum, or hemi diaphragm towards the affected area, and compensatory over inflation in the adjacent non–atelectatic lung
- **Pneumothorax:** air in the pleural space with no vascular bed surrounding the visceral pleura on chest radiograph
- **APACHE (Acute Physiology and Chronic Health Evaluation) II:** a point score ranging from 0-71, calculated from 12 measurements (age, temperature (rectal), mean arterial pressure, pH, heart rate, respiratory rate, sodium (serum), potassium (serum), creatinine, hematocrit, white blood cell count, GCS) higher scores correspond to more severe disease and higher risk of death
- **SAPS (Simplified Acute Physiology Score) II:** point score ranging from 0-163, as APACHE
- **MRC (Medical Research Council):** grades strength in functional muscle groups in each extremity, ranging 0-5, a score of 5 corresponds to normal – healthy strength

APPENDIX II**SCHEME FOR UNASISSTED VENTILATION WITH TRACHEOSTOMY**

The following suggested scheme can be used for unassisted ventilation with a tracheostomy, but should be individualized in every patient:

1. Unassisted ventilation for 30 minutes, three times per day
2. Unassisted ventilation for 1 hour, three times per day
3. Unassisted ventilation for 2 hours, three times per day
4. Unassisted ventilation for 4 hours, three times per day
5. Unassisted ventilation for 6 hours, two times per day
6. Unassisted ventilation for 18 hours
7. Unassisted ventilation for 24 hours